Day 32 - Homework

Friday, May 5, 2023 12:16 PM

Learning Targets	Self-Assessment			
Start by reviewing the Polynomial Growth Check. Flint Period 1: https://app.flintk12.com/activities/polynomial-revi-4e11bf/sessions/new Flint Period 4: https://app.flintk12.com/activities/polynomial-revi-78b8ca/sessions/new	I struggle greatly with this concept. Even if showed an example, I cannot follow the problem.	I struggle a bit with this concept. I would need an example or help from a teammate to complete a problem.	I understand this concept and can solve problems without help. Errors are present.	I fully understand this concept and can give an in-depth answer when solving problems.
Rubric Score	5-6	7	8	9-10
1. I can sketch a graph of a polynomial given an equation in factored form as in problems 8-35, 8-37, 8-106, 8-107, CL 8-179, and CL 8-183. Graphs of polynomials (article) | Khan Academy Khan Academy Graphs of polynomials: Challenge problems (article) | Khan Academy				
2. I can, given a graph or description, write an exact equation of a polynomial function that as in problems 8-48, 8-49, 8-50, 8-54, 8-56, and CL 8180. Khan Academy				
3a). I can sketch a polynomial that has a given number of real and complex roots, or determine the number of real and complex roots from the graph, as in problems 8-104, 8-105, 8170, CL 8182. 3b). I can use the discriminant to determine if a quadratic function has real or complex roots as in problems 8-88, 8-110, 8-172 and CL 8-181				
4. I can simplify, add, subtract, multiply, and divide imaginary numbers, complex numbers, and complex conjugates as in problems 8-70 to 8-73, 8-76, 8-111, 8-141, 8-149, 8-156, and CL 8184.				
5. I can divide polynomials as in problems 8-124, 8-148, 8-175, and CL 8185 Khan Academy Khan Academy				
6a). I can, given roots, find a possible polynomial as in problems 8-87, 8-139, 8-147, 8176, and CL 8187. 6b). I can, given an equation, find all real and complex roots, or all factors, as in problems 8-121, 8-124, 8-143, 8-148, 8-154, and CL 8186.				

Targets #1 + 6b

1. State the degree of each polynomial and then find the exact roots for each equation.

a) $f(x) = 4(x-2)^2 + 3$ b) $f(x) = -2(x-9)^3(x+7)$

c) Sketch the graph of the polynomial function in part b.

8-121.
Now Carlos needs to solve $2x^3 + 3x^2 - 8x + 3 = 0$, but his calculator will still only create a standard graph. He sees that the graph of $y = 2x^3 + 3x^2 - 8x + 3$ crosses the x-axis at x=1. Find all three solutions to the equation. Homework Help

8-123.
Which of the following binomials could be a factor of $x^3 - 9x^2 + 19x + 5$?Explain your reasoning. Homework Help

a. $x - 2$
b. $x - 5$
c. $x + 3$
d. $x + 2$

8-124.
Now divide $x^3 - 9x^2 + 19x + 5$ by the factor that you chose in the preceding problem. If it is a factor, use it and the resulting factor to find all the zeros of the polynomial. If it is not a factor, reconsider your answer to the preceding problem and try a different factor. Homework Help

Targets #1+2
2a). Write the exact equation for a graph that passes through the point (1, 56) and has three roots: (–3, 0) with a multiplicity of 1, (5, 0) with a multiplicity of 1, and a root at (2, 0) with a multiplicity of 2.

2b). Sketch the graph described in part a.

Target #3a
3. Make a sketch of a graph $q(x)$ such that $q(x) = 0$ would have the number and type of solutions indicated below.

a) 5 real and 2 complex b) 4 complex c) 2 complex and 4 real

4. Explain why it is impossible for an odd degree polynomial to have no real roots.

Target #3b

CL 8-181.

Decide if each of the following functions has real or complex roots.

$y = 3x^2 + 5x + 4$ $\qquad$ $y = 3x^2 + 5x - 4$

Target #4

6. Simplify each expression.

a) $(3 + 5i)$ $\qquad$ b) $(3i)(2i)$ $\qquad$ c) multiply $7 + i$ by its complex conjugate (in other words make it a real number)

CL 8-184 Simplify each expression.

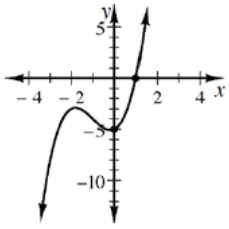
$(3 + 4i) + (7 - 2i)$ $\qquad$ $(3 + 5i)^2$ $\qquad$ $(7 + i)(7 - i)$

$(3i)(2i)^2$ $\qquad$ i^3 $\qquad$ i^{32}

Targets #5+6

7. Use an area model to divide: $(6x^4 + 7x^3 - 36x^2 - 7x + 6) \div (3x - 1)$

8. The graph of $f(x) = x^3 + 3x^2 + x - 5$ is shown below. Use it to determine all real and complex roots.



9. Find a possible quadratic equation in standard form for each pair of roots below. *Use a different method for each part.*

a) $x = 2i, x = -2i$

b) $x = 2 + \sqrt{3}, x = 2 - \sqrt{3}$